Discovering New Mental Diagnoses Through Vectorization of InsaneSpace

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Abstract

As we continue our ~~academic~~ journey in insanity at this institution, we discover how there is a lack of documentation on the sparsity of the conceptual space of mental diagnoses. To combat this issue, we perform computational analysis on InsaneSpace through vectorization of mental symptoms using a machine learning model. After a vectorization to Euclidean space, we then explore two techniques to convert euclidean vectors into InsaneSpace vectors, allowing for analysis of the relationships of mental diagnoses to symptoms and signs.

1 Introduction

InsaneSpace is mostly a nebulous, undefined space, with lots of sparsity and variation through the power set of mental illnesses and disorders. This leads to many psychiatric patients being either not diagnosed or improperly diagnosed [3]. Therefore, to improve accuracy in mental diagnoses, the APA (American Psychiatric Association) and WHO (World Health Organization) created the DSM (Diagnostic and Statistical Manual of Mental Disorders) and ICD (International Classification of Diseases) respectively to help standardize and define diagnostic procedures for basically every mental illness known to medicine [16]. Updates/Patch Notes to the DSM andICD have come out since then, but only continue to expand or further define mental diagnoses and their subtypes. However, these updates tend not to include sparse spaces of lacking knowledge, as the primary purpose of DSM and ICD are as billing and coding manual/classification systems, not as a research review in psychiatric spaces not commonly found in practice. The computational analysis of mental diagnoses, symptoms and signs is likely a first step in recognizing and filling these gaps in diagnostic psychiatry, leading to new mental diagnoses and better recognition of sparse areas of mental diagnostics. We will explore one method of computational analysis of psychiatry in this paper, which is vectorization to better understand the space of mental diagnoses.

2 Motivation

Our motivation for this research model was the fact that our previous publication's definition of "winning" (in the Any% category) and defeating the mental healthcare ~~providers~~ final bosses was too nebulous to practically define a win [27]. Formally, "winning 'the game"' was defined as multiple things, such as having a team of specialized psychiatrists study a winner in a mental asylum/research hospital, or be a foundational case study in psychiatric medicine, or getting a mental disorder named after the winner. However, the authors feel like those former win methods can be influenced by external factors not related to the player's insanity, such as funding and location of the ~~treatment facility~~ dungeon system, and also popularity of the treating physician(s) in academia. Therefore, to ensure equity for all players in this game, we propose a "win" to be a state where a player is in a new, not fully defined subspace in InsaneSpace, potentially creating a new mental disorder/diagnosis.

3 Literature Review

3.1 Undefined Behavior in Psychiatry

Studies on undefined behavior (UB) in psychiatry are hard to find. However, there is one study by Fard et al. [8] that followed 24 adolescent psychiatric patients over seven years that had mental symptoms. Nine of those patients after that time "remained well", as the symptoms were derived from parental conflict. Fifteen of those patients were mentally ill during that period, and 12 of those patients were diagnosable as adults as the "atypical adolescent psychiatric disorder became more typical in adulthood".

3.2 Taxonomy of Mental Disorders

The DSM (Diagnostic and Statistical Manual of Mental Disorders) [2] is the industry standard in the United States for the diagnosis and classification of mental disorders, with a comprehensive list of diagnostic criteria and symptoms for each mental disorder. The DSM has also influenced the development of the mental diagnoses and symptom sections of ICD codes [25]. The DSM is used in conjunction with ICD-10-CM codes (the International Classification of Diseases, Tenth Revision, Clinical Modification) [5], as that is the standard in US healthcare. These codes are also used in the broader healthcare system for simplifying medical documentation regarding symptoms and diagnoses. Each mental diagnosis has at least one ICD-10-CM code, with some diagnoses having multiple due to sub-diagnoses, comorbid symptoms, and/or severity of the disorder. The ICD-10-CM codes associated with mental disorders are prefixed by F, and some mental symptoms are prefixed with R4{0-6}.

The Unified Medical Language System (UMLS), developed by the U.S. National Library of Medicine (NLM), is a comprehensive biomedical terminology integration system designed to improve interoperability between health information systems [15]. Rather than being a single vocabulary, UMLS is a metathesaurus that aggregates and maps concepts from many medical vocabularies and classification systems. UMLS uses Concept Unique Identifiers (CUI), which are stable alphanumeric identifiers assigned to a single biomedical concept, to link together all synonymous terms for that concept across different vocabularies and coding systems. UMLS consists of three primary components: the Metathesaurus (which links synonymous terms and concepts across vocabularies), the Semantic Network (which categorizes concepts into broad semantic types and defines relationships among them), and the SPECIALIST Lexicon and Lexical Tools (which support natural language processing) [26].

3.3 LLM Usage in Medical Contexts

Currently, there are many uses for LLMs (Large Language Models) in the medical field [20] [24]. The one use relevant to our work is LLM data annotation and literature-based knowledge extraction, which was used in Gao et al. [10] in creating a Mental Disorders Knowledge Graph (MDKG). This graph was primarily made through LLMs going through and extracting important data points from literature, from which these data points were then neatly sorted and organized by the LLM. Later, data points were then validated and edited by hand if the phrases were unclear. The authors also propose a way to embed patient data in the MDKG through similarity matching with ICD-10 codes and an embedding model.

Freidel & Schwarz [9] propose the use of knowledge graphs (KGs) in future psychiatric research, in which these KGs could be used with a machine learning algorithm to create vector representations for use in downstream ML applications. For example, Pu et al. used graph embeddings to create links and relations in an Alzheimer's Disease graph [23]. Mal et al. used many word and graph embeddings to semantic similarity matching in the Unified Medical Language System [18]. BioWordVec [28] was found to be the best performing method for sentence embeddings, something recommended by Chanda et al. [6] in definitions of UMLS concepts to improve embedding, allowing for a higher performing semantic search when training data is small.

4 Definitions

We define InsaneSpace, $\boldsymbol{I}$, as a conceptual space [11], where a mental diganosis is a manifold and each dimension is a mental/behavioral symptom. Without comorbidity, InsaneSpace can also be represented as a bipartite graph, where $\boldsymbol{S}$ is the set of mental symptoms and $\boldsymbol{D}$ is the set of mental disorders. However, since only specific subsets/combinations of $\boldsymbol{S}$ are related to certain mental diagnoses (e.g., an ADHD diagnosis requires at least six symptoms of inattention and/or hyperactivity- impulsivity to be diagnosed [2]), we define the disorder-symptom relation as $R \subseteq \boldsymbol{D} \times \mathcal{P}(\boldsymbol{S})$, where any mental disorder maps to a subset of mental symptoms. This relation has an inverse, $R^{-1} \subseteq \mathcal{P}(\boldsymbol{S}) \times \boldsymbol{D}$, where a subset of mental symptoms is related to a mental disorder. This is not a diagnostic relation yet, as patients can have multiple diagnoses from a subset of mental symptoms, requiring a diagnostic relation to be $I \subseteq \mathcal{P}(\boldsymbol{S}) \times \mathcal{P}(\boldsymbol{D})$, where any subset of mental symptoms can be related to any subset of mental diagnoses.

5 Methods

We begin by programmatically extracting data from UMLS and the ICD-10-CM codes (DSM-V Related Codes, primarily F codes). ICD-10-CM code extraction was shockingly simple; see Figure 1.

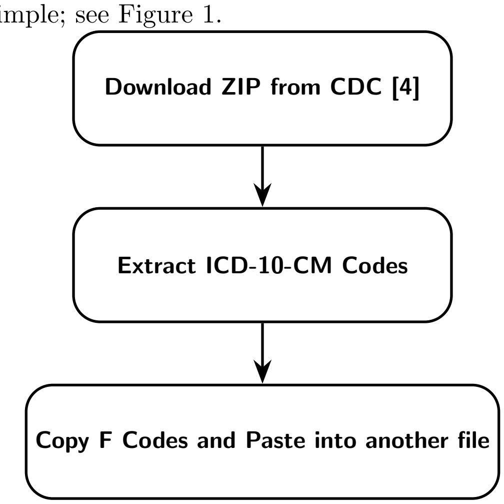
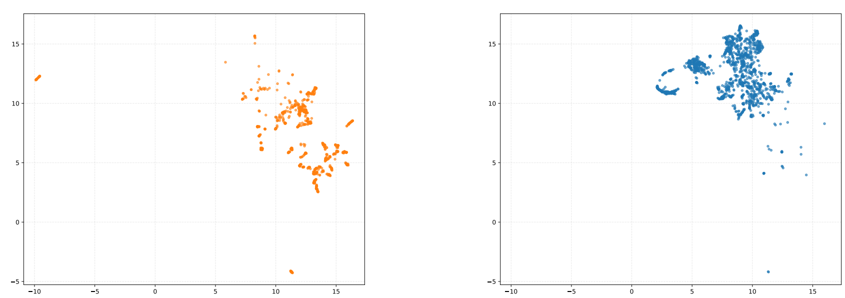


Fig 1. Workflow for ICD-10-CM F Code Extraction

UMLS data extraction was a bit more complicated. We decided to use a vibecoded (yes, we are procrastinating on our procrastination) Python script and the UMLS REST API [22] to retrieve our data. This script started by traversing the DAG (directed acyclic graph [7]) that is the UMLS Concept Map. We started at UMLS CUI: C0237088, Mental and Behavioral Signs and Symptoms [21], and traversed the narrower concepts (for our purposes, narrower categories of mental symptoms or signs or mental symptoms or signs) recursively until we reached CUIs/nodes that had no narrower concepts. To make sure that we didn't duplicate nodes in our traversal, we made sure to keep a list of mental symptoms/signs that we had already visited to check against later queries. Finally, we repeated the traversal process for nodes that returned HTTP errors, as there were often some network issues when running our script. From this script, we were able to find out the whole sub-graph from CUI C0237088. For example,

one fun thing we found was that Bioterrorism is connected downstream from mental symptoms:
(C0001807: Aggressive behavior $\longrightarrow$ C0009671: Conflict (Psychology) $\longrightarrow$ C0042693: Violence$\longrightarrow$ C0039565: Terrorism$\longrightarrow$ Bioterrorism).
We did remove those nodes from our search as the scope of this paper is not related to forensic psychiatry, although future work may find this subgraph interesting (e.g., CUI: C1449772).

After gathering useful node data, we then began the vectorization process using the Bio+Clinical BERT model [1], which was initialized from BioBERT [14] and trained on an additional medical records dataset [13]. Vectors were generated primarily based on the description of Mental Symptoms/Signs (UMLS CUIs). For ICD-10-CM F codes, we used the full string for vectorization. This allows for more accuracy as the additional context provided by the definition or the more specificity provided with an exact code [6].



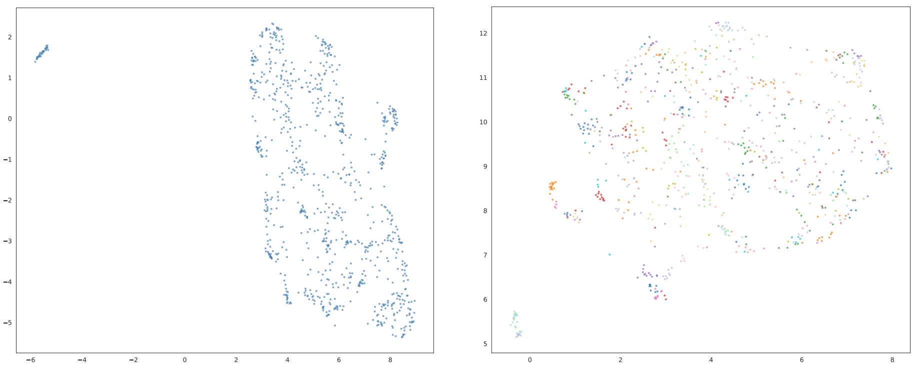
(a) UMAP DSM Projection

(b) UMAP UMLS Projection

Fig 2. UMAP embeddings for DSM and UMLS datasets.

Figure 2 is a visualization of our first step towards modeling InsaneSpace. Using UMAP (Uniform Manifold Approximation and Projection for Dimension Reduction [19]), the embeddings of both the DSM and UMLS items are reduced down to a two-dimensional set of vectors, allowing for visualization on a simple figure as seen here.

We decided to attempt to use 2 methods to transform the DSM and UMLS data. We tried using a cosine similarity (Eq. 1) [17] based method, where every mental diagnosis is transformed into a vector such that each dimension in that vector represented a mental symptom. For example, ICD-10-CM code F90.9 (Attention-deficit hyperactivity disorder (ADHD), unspecified type) and CUI C0424235 (Fidgeting) have a cosine (directional) similarity of roughly 81%. We decided to use min–max normalization to also make the space a bit more evenly distributed, as there was a set of diagnoses unique to a majority InsaneSpace; that is, that set could be removed from the dataset and it would result in a much smaller and tighter InsaneSpace.



(a) Cosine Similarity Vectors Without Min–max Normalization

(b) Cosine Similarity Vectors With Min–max Normalization

Fig 3. UMAP embeddings for DSM cosine similarity vectors in relation to UMLS vectors

$$S_C(A,B) := \cos(\theta) = \frac{\mathbf{A}\cdot\mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\cdot\sqrt{\sum_{i=1}^{n} B_i^2}} \tag{1}$$

The other method used was a linear projection of the DSM set of embedding on to a UMLS subspace in BioBERT's vectorspace. In order to create the subspace, we used the UMLS vectors in BioBERT's vectorspace in order to create a linear map into InsaneSpace, $\mathbb{R}^B \rightarrow \mathbb{R}^I$. We began with defining the matrix $\mathbf{I}$, where the columns are the UMLS set of vectors in $\mathbb{R}^B$. From $\mathbf{I}$, we solved $\mathbf{I}c \approx x$ for all vectors using $\min_c |x - \mathbf{I}c|^2$, which when solved results in c, the vector in InsaneSpace, being transformed via $c = \mathbf{I}^+ x$, where $\mathbf{I}^+ = (\mathbf{I}^T\mathbf{I})^{-1}\mathbf{I}^T$, which is the Moore–Penrose pseudoinverse [12].

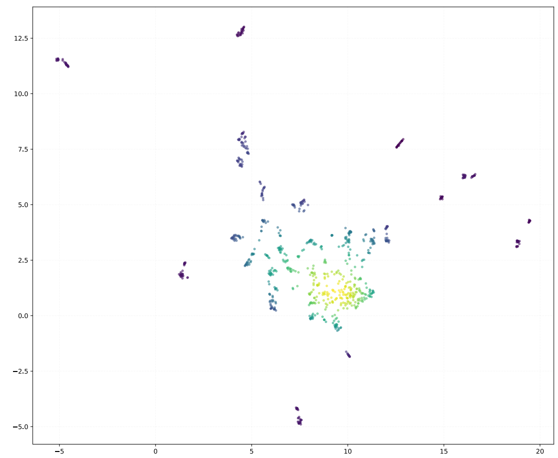


Fig 4. Linear Projection Method of Model Vectorspace to InsaneSpace

Finally, we have the Diagnostic coverage of InsaneSpace which is calculated through the combinations of $\mathcal{P}(\boldsymbol{D})$ and the $\frac{1}{2}\min(\mathrm{dist}(u, v) : u, v \in \boldsymbol{D}, u \neq v)$, leading to the insane amount of quantization "levels" being $\approx 10^{1218.23}$.

6 Results and Discussion

Our work in collecting the inital training data for this research could have been better. Better clustering or vectorization could have likely been achieved through adding descriptions to the ICD-10-CM F codes through LLM traversal of the DSM-5-TR; we attempted this method but realized we could either not afford the compute or the time to finish this work before Sigbovik 2026. We also did not strictly verify that the UMLS CUIs were a symptom/sign type CUI; this was hard to do as there are multiple categories of CUIs that could have fit or not fit the discription we were looking for in CUIs. For example, the "Mental or Behavioral Dysfunction" semantic type was the one type we were primarily using, but there could have been other types or CUIs that were relevant for our work or CUIs in this the "Mental or Behavioral Dysfunction" semantic type that were not relevant to our work. Finally, the UMLS Metathesaurus could have been traversed without an API; however, we also do not have the money for a Oracle database server licence or the compute required to run complex graph/database traversal queries.

The cosine similarity method does not seem useful for quantifying InsaneSpace for the purposes of empty space, as the vectors it produced seems to filled InsaneSpace well. We mentioned a group of diagnoses that have added more magnitude to the symptom similarity; this expanded InsaneSpace dramatically, but we did not feel as if this model was correct, as the extra InsaneSpace in relation to the magnitude of the diagnosis vectors was too high. Attempting to fix this issue through min-max normalization led to other issues with similar diagnoses not properly being clustered. Furthermore, these vectors were a majority high in magnitude, as trivally many, if not all, mental diagnoses have a cosine similarity high in relation to mental symptoms and signs. Therefore, this vector set and method may be useful for comorbidity findings, as the the related symptoms are close to each other and by extension the mental diagnoses have related or share symptoms, meaning they are likely comorbid.

The linear projection method is likely the best one for this use, as the projection more properly tightly clusters similar diagnoses and provides a proper projection method for work such as single symptom linear correlation. However, the projection vectors were not orthonormal, leading to the usage of the Moore-Penrose pseudoinverse rather than a Gram-Schmidt process. The subspace defined by the UMLS mental symptoms was likely also overdefined, as the mental symptoms were not linearly independent. This means an improvement would be to work on manually pruning similar or extremely semantically similar mental symptoms could improve the diagnosis vectors in InsaneSpace.

Finally, the high dimensional defintion of InsaneSpace lead to a lot of quantization levels, or "boxes" where only one mental diagnosis can exist, means that practically InsaneSpace has a lot of discoveries left to do. However, our quantization metric might be off as the diagnoses we used were often subtypes or even subtypes with specific expressions of certain related symptoms within a diagnoses. This means that InsaneSpace could be a lot less undiscovered than expected; however, note that some new mental diagnoses stem from subtypes that are prominent and distinct enough to create a new diagnosis, meaning that a single covering at this quantization level could be a still incomplete InsaneSpace.

7 Conclusion

InsaneSpace is still a sparse, nebulous space. However, with our work, it likely is slightly more defined in parts where there is a lack of mental diagnoses. We hope it will lead to new and more equitable game wins through our new definition of an Any% win. InsaneSpace seems to be very empty in our quantized model, leading us to believe that updates to the mental healthcare RPG with this model will be extremely easy to win before balance updates and emergency nerfs to the emptiness of this model. Overall, more work needs to be done on this win model, including beta testing before rolling out to main.

8 Ethics Disclosure

AI was obviously used in the production of this paper.

9 Not for Clinical Use

The authors are clearly not the physicians nor the heathcare professionals nor the staff at this institution, so please do not use our work to invent new mental illnesses for us to catch. We are currently working on the 100% completion category, so please don't slow us down through the release of new mental diagnoses to obtain.

9.1 Disclaimer

The statements in this publication have not been evaluated by the Food and Drug Administration. This product is not intended to diagnose, treat, cure, or prevent any disease.

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